

Notes: Campbell and Cocco (JF, 2015)

A Model of Mortgage Default

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1 Big picture

- **Question.** What determines household mortgage default once we allow for realistic labor income risk, house price risk, inflation risk, interest rate risk, and endogenous mortgage pricing by competitive lenders?
- **Setting.** This is a structural household-finance paper. A life-cycle household buys a fixed house, chooses consumption and savings, and each period can continue paying, default, prepay/cash out, or (for FRMs) refinance.
- **Core challenge.** Mortgage default is not driven by negative equity alone. A borrower may be underwater but still keep paying if liquid resources are sufficient and future option value is high. Conversely, tight borrowing constraints can trigger default at less negative equity.
- **Main conceptual contribution.** The paper combines a dynamic mortgage-choice/default model with *equilibrium mortgage premia*. Lenders price ARMs and FRMs using a risk-adjusted zero-profit condition, so contract pricing and borrower behavior are jointly determined.
- **Main economic mechanism.** Default has a “dual trigger”:
 - a **wealth trigger**: sufficiently negative home equity,
 - a **liquidity trigger**: high mortgage-payment-to-income pressure and low cash-on-hand.
- **ARM versus FRM result.** For moderate house-price declines, ARM defaults tend to occur when interest rates rise because payments reset upward, while FRM defaults are more likely when rates fall and underwater borrowers cannot refinance out of an expensive fixed-rate loan.
- **Affordability result.** LTV and LTI matter through different channels. Higher LTV raises the probability of negative equity; higher LTI raises the chance that borrowing constraints bind and therefore makes default optimal at less negative equity.
- **Pricing result.** FRM premia rise strongly with initial interest rates because the borrower refinance option becomes more valuable. ARM premia are much flatter and are only slightly increasing in the baseline model.
- **Selection result.** The model can explain why ARMs performed worse in the housing bust if riskier borrowers – especially borrowers whose income is more exposed to recessions/interest-rate states – select into ARMs.
- **Main welfare result.** When initial rates are low, FRMs are slightly preferred because borrowers can lock in cheap financing. When initial rates are high, ARMs become more attractive because FRM premia rise sharply.
- **Broader takeaway.** Mortgage market risk is not just about whether rates float or not; it is about how contract design interacts with borrower heterogeneity, home equity, liquidity constraints, and lender pricing.

2 Data and measurement (key objects)

2.1 Why this is a structural model paper rather than a reduced-form empirical paper

- The paper does **not** estimate a causal treatment effect with cross-sectional variation, an IV, or a difference-in-differences design.
- Instead, it calibrates a dynamic model using external data and studies the model's implied behavior under alternative contracts and aggregate states.
- The “measured objects” are therefore:
 - the stochastic processes for inflation, real rates, labor income, and house prices,
 - mortgage ratios such as LTV, LTI, and MTI,
 - model-implied default, cash-out, refinance, and mortgage-premium probabilities,
 - and comparisons between model outcomes and U.S. mortgage-market data.

Interpretation. The paper is asking whether a disciplined structural model can explain both mortgage default behavior and mortgage pricing patterns, not whether a particular observed policy caused default to rise or fall.

2.2 Household preferences and horizon

The household solves a finite-horizon life-cycle problem with utility

$$E \left[\sum_{t=1}^T \beta_i^{t-1} \frac{C_{it}^{1-\gamma_i}}{1-\gamma_i} + \beta_i^T b \frac{W_{i,T+1}^{1-\gamma_i}}{1-\gamma_i} \right].$$

- C_{it} is nondurable consumption.
- $W_{i,T+1}$ is terminal real wealth, including financial and housing wealth when the household still owns the house.
- The baseline calibration uses annual periods, initial age 20, terminal age 50, discount factor $\beta = 0.98$, risk aversion $\gamma = 2$, and a bequest motive parameter $b = 400$.
- Housing services are fixed because the paper takes house size H_i as given once the mortgage is originated.

Interpretation. The model isolates mortgage decisions by abstracting from ongoing housing adjustment. This keeps the state space manageable and lets the paper focus on default, cash-out, and refinance decisions.

2.3 Aggregate state variables: inflation, real rates, and house prices

Expected inflation follows

$$\pi_t = \mu_\pi(1 - \phi_\pi) + \phi_\pi \pi_{t-1} + \varepsilon_t,$$

and the ex ante real short rate follows

$$r_t^\ell = \mu_r(1 - \phi_r) + \phi_r r_{t-1}^\ell + \xi_t.$$

The one-year nominal yield is

$$y_{1t} = r_t^\ell + \pi_t.$$

Real house-price growth is

$$\Delta p_t^H = g + \delta_t.$$

- Inflation expectations are calibrated from the Livingston survey, 1987–2012.
- One-year nominal yields come from the Federal Reserve; the real rate is constructed as nominal yield minus expected inflation.
- House-price moments come from PSID data and the Case-Shiller 10-City index.
- The baseline calibration sets mean log real house-price growth to 0.003 and the standard deviation to 0.162.

Interpretation. Inflation and real interest rates matter separately because nominal mortgage balances, nominal payments, and the real burden of debt all depend on them. House prices matter because they determine home equity.

2.4 Labor income process

Log real labor income is

$$\ell_{it} = f(t, Z_{it}) + \nu_{it} + \omega_{it},$$

with a persistent component

$$\nu_{it} = \nu_{i,t-1} + \eta_{it}.$$

- $f(t, Z_{it})$ is a deterministic age/profile term estimated from PSID data.
- η_{it} is a permanent income shock and ω_{it} is a transitory shock.
- The baseline standard deviations are $\sigma_\eta = 0.063$ and $\sigma_\omega = 0.225$.
- The paper also studies higher transitory income risk ($\sigma_\omega = 0.35$) and a case where income realizations are correlated with interest-rate states.

Interpretation. Income risk matters for two reasons. It shapes precautionary saving ex ante, and it determines whether mortgage payments become unaffordable ex post when bad shocks arrive.

2.5 Mortgage origination ratios and their meanings

The initial loan amount is

$$D_{i1} = (1 - d_i)P_1 P_1^H H_i,$$

so the down payment ratio is d_i . The main origination ratios are

$$LTV_i = 1 - d_i, \quad LTI_i = \frac{(1 - d_i)H_i}{L_{i1}}, \quad MTI_i = \frac{M_i}{L_{i1}}.$$

- **LTV** measures initial leverage and therefore the starting equity cushion.
- **LTI** measures how large the debt is relative to income.

- **MTI** measures payment affordability, that is, the direct cash-flow burden of the mortgage.

The baseline contract has **LTV** = **0.90** and **LTI** = **4.5**.

Interpretation. The paper's key policy message is that LTV and affordability ratios are not substitutes. They operate through different margins and should be considered jointly.

2.6 FRM and ARM payment formulas

For a fixed-rate mortgage, the real payment is a standard annuity payment:

$$M_{it}^{FRM} = \frac{D_{i1}}{P_t} \cdot \frac{Y_i^{FRM}(1 + Y_i^{FRM})^T}{(1 + Y_i^{FRM})^T - 1}.$$

For an adjustable-rate mortgage,

$$M_{it}^{ARM} = \frac{y_{it}^{ARM} D_{it}^{ARM} + \Delta D_{it}^{ARM}}{P_t}, \quad y_{it}^{ARM} = y_{1t} + \Psi_i^{ARM}.$$

- Y_i^{FRM} is the fixed FRM rate plus a premium.
- y_{it}^{ARM} is the short rate plus an ARM premium.
- For FRMs, borrowers can refinance when rates fall, paying a refinancing cost.
- The paper also studies an ARM teaser contract with a temporarily low initial premium.

Interpretation. The FRM embeds a valuable borrower refinance option. The ARM instead exposes the borrower to payment resets. Those contract differences drive both default patterns and lender premia.

2.7 Default, renting, and household fallback after default

If the borrower defaults, the lender seizes the house and the borrower moves permanently into rental housing. Rental cost equals a user-cost formula:

$$u_{it} = \left[y_{1t} - E_t \left(e^{\Delta p_{t+1}^H + \pi_t} - 1 \right) + \tau_p + m_p \right] P_t^H H_i.$$

- Mortgages are effectively nonrecourse in the baseline model: the lender cannot seize financial savings or future labor income.
- Default excludes the borrower from future credit-market access and forces renting.
- The model also includes an exogenous moving probability; if a household must move while underwater, default occurs.

Interpretation. Default is costly and irreversible, which is why households do not walk away as soon as equity turns negative.

2.8 Lender cash flows, profitability, and mortgage premia

If the borrower keeps paying, the lender receives scheduled mortgage payments. If default occurs, the lender gets the house net of foreclosure loss. If prepayment or FRM refinancing occurs, the lender receives principal back. The model summarizes risk-adjusted profitability as

$$PR_{ij}(S) = \frac{\sum_t PV_i[CF_{ijt}](S_1, \dots, S_t)}{(1 - d_i)H_i}, \quad PR_{ij}(S_1) = E_i[PR_{ij}(S)].$$

- Mortgage premia Ψ_i^{ARM} and Ψ_i^{FRM} are chosen so lenders achieve a target risk-adjusted profitability.
- The paper uses **10%** as the benchmark profitability target and also studies 8% as a more competitive environment.

Interpretation. The premium is not an arbitrary spread. It is an equilibrium object pinned down by borrower behavior and lender pricing.

2.9 Current-state default objects

Two objects summarize the default state of a mortgage:

$$LTV_{ijt} = \frac{D_{ijt}}{P_t P_t^H H_i}, \quad MTI_{ijt} = \frac{M_{ijt}}{L_{it}}.$$

The paper also decomposes default as

$$Pr(\text{Default}) = Pr(\text{Equity} < 0) \times Pr(\text{Default} \mid \text{Equity} < 0).$$

Interpretation. LTV tells us whether default is even “in the money.” MTI tells us how painful it is to keep paying. The entire dual-trigger logic of the paper sits inside these two variables.

2.10 Calibration and simulation design

- **Inflation and rates:** Livingston survey plus Federal Reserve yields.
- **Labor income:** PSID, 1970–2005, with age profiles estimated by education group.
- **House prices:** PSID and Case-Shiller moments.
- **Mortgage-market facts for comparison:** FHFA Monthly Interest Rate Survey, Freddie Mac cash-out refinance data, Mortgage Bankers Association delinquency/foreclosure data.
- **Simulation:** 800 aggregate paths and 50 individuals per path, so 40,000 household-path observations for each contract/initial-state/borrower type combination.

Interpretation. The model is disciplined by real data, but its quantitative results come from solving and simulating the structural problem rather than estimating a reduced-form regression.

3 Models and quantitative specifications (all equations + interpretation)

3.1 Household dynamic problem

Each period, a homeowner chooses:

- nondurable consumption,
- whether to default,
- whether to prepay/cash out,
- and for FRMs, whether to refinance.

The relevant state variables include age, cash-on-hand, the inflation rate, the real interest rate, the real house-price level, the persistent component of income, and mortgage-status indicators. For FRMs there is an additional state variable indicating whether the mortgage has already been refinanced.

Interpretation. This is a fully dynamic option problem. Borrowers are not deciding whether to default in isolation; they are comparing the value of continuing, defaulting, refinancing, and cashing out, taking future uncertainty into account.

3.2 The dual-trigger default mechanism

The paper's central mechanism is:

- **Negative equity** raises the value of default because the borrower can hand the house to the lender and walk away from an underwater position.
- **Binding liquidity constraints** raise the value of default because default immediately removes mortgage payments and relaxes cash-flow pressure.

This means:

- a borrower may *not* default even with negative equity if savings are high and cash-flow pressure is manageable,
- but a borrower may default at a relatively mild negative-equity level if MTI is high, savings are low, and income is weak.

Interpretation. This is the main difference from a pure option-theoretic “equity-only” story. Liquidity matters because households are optimizing over consumption and savings, not just mortgage status.

3.3 Why ARMs and FRMs fail in different states

- **ARM default risk** is mostly a **cash-flow risk**. When rates rise, ARM payments rise, MTI goes up, and constrained borrowers may default even before they are deeply underwater.

- **FRM default risk** is more of a **wealth/refinancing risk**. When a borrower locked in a high FRM rate and later house prices fall, the borrower may be unable to refinance just when rates drop. That creates default in low-rate states with negative equity.

Interpretation. The same aggregate house-price decline has different consequences depending on contract structure. That is why the paper emphasizes contract-specific default triggers rather than a generic “all mortgages are risky in bad housing states” conclusion.

3.4 Unconditional quantitative exercises

The unconditional exercises average over simulated aggregate states and idiosyncratic labor-income shocks. These experiments produce:

- average default probabilities,
- average cash-out rates,
- FRM refinance rates,
- mortgage premia,
- lender profitability by termination state,
- and borrower welfare comparisons across contracts.

The main comparative statics vary:

- initial interest rates,
- LTV and LTI at origination,
- labor-income risk,
- correlation of income with interest-rate states,
- refinancing inertia,
- teaser-rate design,
- target lender profitability.

Interpretation. These are not one-off simulations; they are a structured battery of counterfactuals that let the paper separate wealth, liquidity, contract-design, and borrower-selection channels.

3.5 Conditional aggregate-state exercises

The paper then looks at *conditional* default rates: how mortgage outcomes differ across specific simulated aggregate paths. This has two purposes:

- identify which aggregate states generate default waves,
- compare the model to the U.S. path during the 2005–2010 housing bust.

A default wave is associated with:

- very large house-price declines,

- for ARMs: often an increase in interest rates,
- for FRMs: often a decrease in rates when borrowers are trapped in expensive fixed-rate contracts by negative equity.

Interpretation. Looking only at average default probabilities hides the fact that the same average can come from very different macro states.

3.6 Welfare metric

Contract comparisons are reported in **consumption-equivalent variation**. A positive welfare gain of ARM means the borrower would be willing to give up that percentage of a constant consumption stream to have the ARM rather than the FRM.

Interpretation. This translates complex dynamic differences into a single economically meaningful object: which contract the borrower prefers ex ante.

4 Identification and credibility (what makes the quantitative exercise persuasive)

4.1 There is no IV or quasi-experiment here

- The paper does not identify a causal effect using external variation.
- There is also no structural estimation of the full model on loan-level mortgage data.
- Instead, credibility comes from internally coherent optimization, externally disciplined calibration, and multiple margins of comparison with the data.

4.2 Why the exercise is still credible and useful

- **Micro-founded pricing.** Mortgage premia are endogenized rather than imposed. That makes it possible to study how borrower behavior feeds back into mortgage rates.
- **Rich household heterogeneity.** The model allows income risk, interest-rate exposure, leverage, and discounting to vary, which is essential for mortgage selection.
- **Mechanism separation.** The model isolates three channels that are hard to disentangle in raw data: negative equity, payment affordability, and contract-specific option value.
- **External validation.** The paper compares the model to observed mortgage premia, ARM market shares, cash-out activity, refinancing patterns, and foreclosure dynamics during the housing crisis.

Interpretation. The model is most persuasive not because it nails every data moment, but because it organizes many distinct facts with one coherent mechanism.

4.3 Where the model fits the data well

- Borrowers do not default immediately at small negative equity.
- Cash-out activity rises in house-price booms.
- ARM usage is more attractive when rates are high.
- Higher ARM default during the crisis can be rationalized when ARM borrowers are riskier or more cyclically exposed.
- FRM premia rise with interest rates, especially once the refinance option is recognized.

4.4 Where the model struggles quantitatively

- The model has trouble matching the very high ARM premia observed in the data when interest rates are low.
- Baseline FRM premia rise too steeply with the initial level of rates unless refinancing inertia is introduced.
- The paper must appeal to borrower selection and/or changes in lender competition to better match the full time-series behavior of premia.

Interpretation. The paper is strongest on mechanisms and qualitative comparative statics. Some quantitative asset-pricing moments remain hard to fit exactly.

4.5 Main modeling limitations

- House size is fixed after origination.
- Default is effectively nonrecourse in the baseline model.
- Cash-out is modeled mainly through home sale rather than second liens or HELOCs.
- No unsecured debt, bankruptcy choice, or risky portfolio choice.
- Aggregate house-price risk is central; richer local/idiosyncratic housing dynamics are abstracted from.

Interpretation. None of these limitations overturn the main dual-trigger logic, but they matter for precision and for future empirical implementation.

5 Tables and figures

5.1 Figure 1. LTV, MTI, and LTI over time for newly originated U.S. mortgages

Read.

- The striking movement is in **LTI**: it rises strongly in the 2000s.
- **MTI** does not rise nearly as much because nominal mortgage rates were low.
- **LTV** on first liens does not look extraordinarily high by historical standards.

Economic message. The pre-crisis deterioration in mortgage affordability may have shown up more in *income-based leverage* than in first-lien LTV. This motivates the paper's emphasis on LTI/MTI in addition to equity.

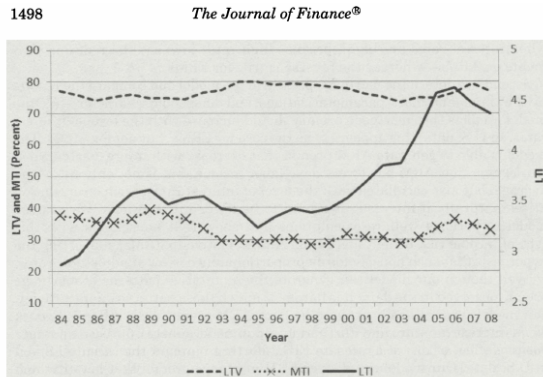


Figure 1. LTV, MTI, and LTI over time for the United States. LTV data are from the MIRS of the FHFA. LTI are calculated as the ratio of the average loan amount obtained from the same survey to the median U.S. household income obtained from Census data. MTI are calculated using the same income measure and the loan amount, maturity, and mortgage interest rate data from the MIRS.

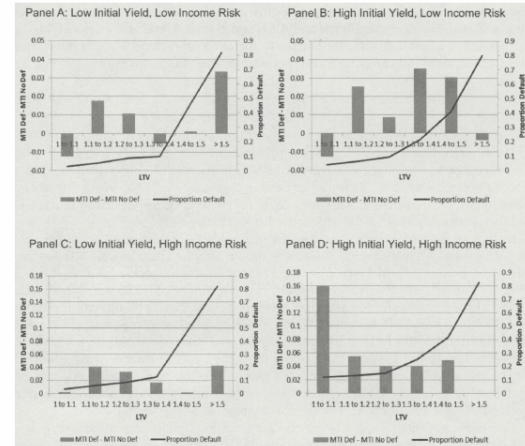


Figure 2. Difference in MTI between households that default and households that do not default and the proportion of defaults as a function of home equity for the ARM contract. The data are obtained by simulating the model for the ARM contract. Low (high) income risk refers to households with a standard deviation of temporary income shocks equal to 0.225 (0.35). LTV is the ratio of current house value to the outstanding loan principal. The vertical bars plot, for each level of negative home equity, MTI for those households that choose to default minus MTI for those households that choose not to default.

5.2 Figure 2. ARM default as a function of negative equity and MTI pressure

Read.

- Default probabilities remain low for mild negative equity and rise only when borrowers become substantially underwater.
- The bars show that defaulting households tend to have higher MTI than otherwise similar households that do not default.
- The MTI difference is especially large in the high-rate, high-income-risk panel.

Economic message. This is the cleanest exhibit for the paper's dual-trigger theory: negative equity matters, but cash-flow stress is what pushes constrained borrowers over the edge.

5.3 Table IV. Initial rates, mortgage premia, default, and welfare

Read.

- ARM premia barely move: about 1.50% at low initial rates and 1.60% at high initial rates.

- FRM premia rise sharply: the annuity-relative FRM premium goes from 1.69% to 2.63%.
- ARM default probability falls slightly as initial rates rise (0.044 to 0.037), while FRM default probability rises (0.034 to 0.051).
- The welfare gain of ARM relative to FRM is about -0.12% at low rates and $+1.12\%$ at high rates.

Economic message. FRMs are appealing when rates are already low because borrowers can lock them in cheaply. When rates are high, the refinance option becomes expensive for lenders to provide, so FRMs require much larger premia and ARMs become more attractive.

Table IV
Probabilities of Default, Cash-Out Prepayment, and Interest-Rate Refinancing, Lender Profitability, and Mortgage Premia, Conditional on Initial Interest Rates

This table reports results for mortgage contracts with LTV=0.9 and LTI=4.5 and for a standard deviation of temporary income shocks equal to 0.225, for low and high initial one-year bond yields. Low (High) initial yield corresponds to the state with the lowest (second highest) level of interest rates in our model. For each of these levels, and for both ARM and FRM contracts, the table reports the mortgage premium required by the lender, the initial mortgage payments relative to income, the probability of default, and the probability of cash-out prepayment. For the FRM contract, the table also reports the probability of interest rate refinancing. This table reports probabilities calculated across aggregate states and individual shocks. The table also reports lenders' average profitability as a function of households' decisions. Profitability is calculated as the present discounted value of the cash flows that lenders receive divided by the initial loan amount. The last row reports the welfare gains of ARMs relative to FRMs, under consumption-equivalent variations. The table reports the percentage difference in the constant consumption stream that makes the individual as well off in the ARM contract as in the FRM contract.

Initial One-Year Bond Yield	Low	High
Panel A: ARM		
Prem over one-year bond yield	1.50%	1.60%
Initial mort payment/Inc	0.234	0.410
Prob(Default)	0.044	0.037
Prob(Cash-out)	0.583	0.595
Profitability(Default)	-0.175	-0.139
Profitability(Cash-out)	0.079	0.077
Profitability(Other)	0.162	0.160
Panel B: FRM		
Prem over 20-year bond yield	0.75%	2.85%
Prem over 20-year annuity yield	1.69%	2.63%
Initial mort payment/Inc	0.344	0.433
Prob(Default)	0.034	0.051
Prob(Cash-out)	0.572	0.369
Prob(Refinancing)	0.000	0.471
Profitability(Default)	-0.122	-0.094
Profitability(Cash-out)	0.086	0.077
Profitability(Refinancing)	0.000	0.122
Profitability(Other)	0.137	0.166
Panel C: ARM/FRM		
Welfare gain of ARM	-0.12%	1.12%

B. Mortgage Premia and Profitability

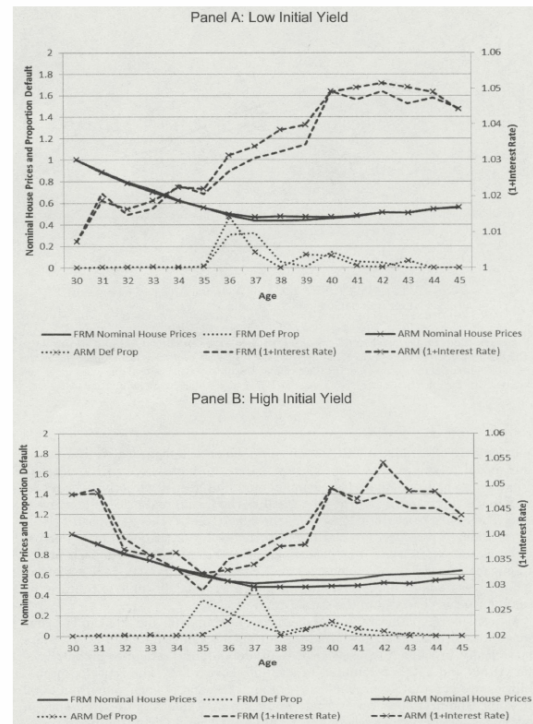


Figure 4. Aggregate characteristics of default waves. This figure plots nominal house prices, nominal interest rates, and the proportion of defaults for aggregate states with a default rate over 80% by mortgage type. Low (high) initial yield refers to the initial interest rates. The data are

5.4 Figure 4. What aggregate default waves look like

Read.

- Extreme default waves are associated with very large house-price declines.
- For low initial yields, ARM default waves coincide with higher subsequent interest rates.
- For high initial yields, FRM default waves tend to occur when rates fall and borrowers are stuck with expensive fixed-rate debt.

Economic message. The same housing bust does not hit all contracts the same way. ARM waves are triggered more by cash-flow resets; FRM waves are triggered more by the interaction of negative equity and trapped refinancing.

5.5 Figure 5 and Table X. A crisis path that mimics the recent U.S. experience



Figure 5. Real house prices and nominal interest rates in the model and in the data. This figure plots real house prices and nominal interest rates over time in the data and in the model. House price data come from the Case-Shiller US 10-City Composite Home Price Index and interest rate data are from the Federal Reserve. House prices in 2005 are normalized to one.

Table X
Cumulative Probabilities of Default, Cash-Out Prepayment, and Refinancing, by Mortgage Type and Income Risk Characteristics, for a Path of the Aggregate Variables that Matches the U.S. Experience
This table reports the probabilities of default, cash-out, and refinancing for a given path of the aggregate variables that matches the U.S. historical experience. The table reports results for LTI=4.5 and LTV = 0.90, for different household income risk characteristics, and for mortgages that began in 2005 and in 2006.

		Year mort 2005		Year mort 2006
Probability through end-2006				
ARM		Cash-out	Default	Default
Base case		0.078	0.000	0.008
Higher inc risk		0.106	0.000	0.008
Correl inc risk		0.078	0.000	0.008
FRM	Refinance	Cash-out	Default	Default
Base case	0.000	0.078	0.000	0.008
Higher inc risk	0.000	0.164	0.000	0.008
Correl inc risk	0.000	0.078	0.000	0.008
Probability through end-2007				
ARM		Cash-out	Default	Default
Base case		0.115	0.000	0.016
Higher inc risk		0.159	0.000	0.060
Correl inc risk		0.216	0.000	0.016
FRM	Refinance	Cash-out	Default	Default
Base case	0.755	0.115	0.000	0.016
Higher inc risk	0.677	0.198	0.000	0.060
Correl inc risk	0.000	0.166	0.000	0.278
Probability through end-2008				
ARM		Cash-out	Default	Default
Base case		0.115	0.008	0.023
Higher inc risk		0.159	0.007	0.067
Correl inc risk		0.216	0.030	0.023
FRM	Refinance	Cash-out	Default	Default
Base case	0.755	0.115	0.001	0.023
Higher inc risk	0.677	0.198	0.009	0.067
Correl inc risk	0.000	0.166	0.246	0.937
Probability through end-2009				
ARM		Cash-out	Default	Default
Base case		0.115	0.015	0.032
Higher inc risk		0.159	0.014	0.075
Correl inc risk		0.216	0.044	0.032
FRM	Refinance	Cash-out	Default	Default
Base case	0.755	0.115	0.002	0.032
Higher inc risk	0.677	0.198	0.010	0.087
Correl inc risk	0.000	0.166	0.289	1.000

Read.

- The model is fed a path with falling real house prices after 2006 and very low rates after 2007.
- For mortgages originated in 2005, there is substantial cash-out before the crash intensifies; for 2006 originations, there is much less cash-out and more later default.
- Under higher or interest-rate-correlated income risk, ARM borrowers show noticeably higher cumulative default probabilities than FRM borrowers.

Economic message. The paper's explanation for the crisis is not simply "ARMs are bad." Rather, ARMs perform worse in the downturn when the borrowers who chose them are more fragile and more cyclically exposed.

5.6 Table XI. Mortgage premia in the data and in the model

Read.

- In the data, ARM premia decline sharply as one-year yields rise: from 3.80% at the lowest-rate bin to 1.27% at the highest-rate bin.
- In the data, FRM premia are much flatter and rise only modestly with rates.

- In the model, FRM premia rise in the right direction, but too steeply in the baseline case.
- The model can move closer to the data for ARMs only when borrower composition or lender profitability varies with the rate environment.

Economic message. The model matches the qualitative FRM pattern better than the ARM pattern. This is where the paper leans on borrower selection and time-varying competitiveness of mortgage supply.

Table XI
Mortgage Premia in the Data and in the Model

Panels A and B report the average ARM and FRM premia in the data for different levels of the one-year bond yield and the slope of the term structure of interest rates. The data are from the Monthly Interest Rate Survey from 1986.01 to 2008.10. Panels C (and D) report some of the model-predicted mortgage premia for ARMs and FRMs, respectively.

One-Year Bond Yield	<0.021	0.021 to 0.041	0.041 to 0.061	>0.061
Panel A: Premium in the Data, Based on One-Year Bond Yield				
ARM premium	3.80%	2.59%	1.55%	1.27%
FRM premium	1.58%	1.53%	1.55%	1.89%
Term structure slope	> 0.016	0.0 to 0.016	-0.016 to 0	<-0.016
Panel B: Premium in the Data, Based on Slope of Term Structure				
ARM premium	2.78%	1.45%	0.82%	
FRM premium	1.56%	1.71%	2.09%	
Panel C: ARM Premium in the Model				
Baseline	1.50%	1.60%	1.60%	1.70%
Lower LTV	1.45%	1.55%	1.55%	1.75%
Higher inc risk	1.55%	1.65%	1.75%	1.95%
Correlated inc risk	1.85%	1.70%	1.75%	1.65%
Lower profitability	1.25%	1.30%	1.30%	1.40%
Panel D: FRM Premium in the Model				
Baseline	1.69%	2.26%	2.63%	4.69%
Lower LTV	1.64%	1.72%	2.28%	3.39%
Inertia = 0.5	1.69%	1.66%	2.18%	2.99%
Lower profitability	1.44%	1.71%	2.18%	3.59%

one-year bond yields, which is reported in the first line of the panel, and then

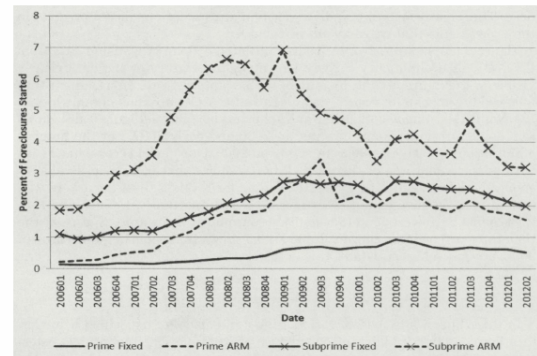


Figure 9. Foreclosures started by loan type (percent). The data are from the National Delinquency Survey of the Mortgage Bankers Association.

5.7 Figure 9. Foreclosures by loan type in the crisis

Read.

- Subprime ARM foreclosures rise the most during the crisis.
- Prime fixed-rate mortgages perform best.
- ARM foreclosures are already elevated before the full peak of the crisis and remain above FRM foreclosures for comparable credit segments.

Economic message. The crisis data are broadly consistent with the paper's message that contract choice and borrower selection interact. A representative-borrower comparison is not enough.

5.8 Other important results not shown above

- **Tables II and III.** Defaulting borrowers have more negative equity, lower cash-on-hand, and higher MTI than observationally similar households who do not default. This is true for both ARMs and FRMs.
- **Table V.** Adding FRM refinancing inertia lowers FRM premia and improves the model's fit to observed FRM spreads.

- **Table VI.** Lower LTI mainly works through affordability and matters relatively more for ARM default; lower LTV mainly works through equity and matters relatively more for FRM default.
- **Table VII.** Higher income risk raises default and premia for both contracts, but especially increases the attractiveness of ARMs for borrowers whose income is positively related to interest-rate states.
- **Table VIII.** Faster expected income growth modestly improves affordability; lower patience raises default and cash-out; adding a stigma cost to default substantially reduces default probabilities.
- **Table IX.** ARM teaser rates increase later premia and slightly raise default; lower target lender profitability compresses spreads and can help rationalize lower observed ARM premia.

6 Short summary

- The paper builds a dynamic life-cycle model of mortgage default with labor income, house prices, inflation, and real interest rates.
- Default is governed by a dual trigger: negative equity plus tight liquidity/affordability conditions.
- ARMs and FRMs fail in different states: ARM default is more about payment resets; FRM default is more about being trapped in an expensive contract when negative equity blocks refinancing.
- LTV and LTI matter through different channels, so regulators and lenders should care about leverage and affordability jointly.
- The model qualitatively matches several mortgage-market facts, but some pricing moments – especially the ARM premium pattern – require borrower selection or time-varying lender competition to fit better.

7 Conclusion

7.1 Core conclusion

The paper shows that mortgage default is best understood as the outcome of a dynamic household optimization problem in which home equity, cash-flow pressure, contract type, and endogenous mortgage pricing all matter simultaneously. Negative equity is important, but it is not sufficient by itself.

7.2 Economic intuition

A borrower keeps paying not because the mortgage is “good” in an absolute sense, but because the value of preserving housing, future credit access, and the option to wait can exceed the short-run gain from default. Default becomes optimal when the mortgage is both underwater enough and painful enough to keep servicing. ARMs and FRMs change which states make that trade-off break.

7.3 Takeaway

The most useful lesson from the paper is that mortgage risk should be analyzed at the level of **contract design + borrower type + macro state**. Looking only at contract labels or only at equity misses the central economics of mortgage default.